

Tom (Pham Hoang Duy) Ngo

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EDUCATION

Santa Clara University

Master of Sciences in Artificial Intelligence

GPA: 3.85

December 2027

University of California, Davis

Bachelor of Arts in Economics

Major GPA: 3.60

June 2025

TECHNICAL SKILLS

Analytics: Machine Learning, AI, Reinforcement Learning, Deep Learning, Neural Network, Predictive Modeling, LLMs

Tech: Python, C/C++, R, BASH, SQL, Linux, scikit-learn, TensorFlow, PyTorch, Airflow, Docker, Git, NS-3, DQN

EXPERIENCE

Research Assistant

Wireless Intelligent Networks (WIN) Lab

January 2026 – Current

Santa Clara, CA

- Developed NS-3 simulation environment to design, train, and evaluate reinforcement learning DQN agents for intelligent WiFi 7 networks optimization, resulting in 300% throughput improvement
- Mentored and instructed lab members on designing, implementing, and executing NS-3 simulations, improving team productivity and ensuring correct and efficient use of the simulation framework

Data Science Intern

Sacramento Municipal Utility Districts (SMUD)

August 2024 – August 2025

Sacramento, CA

- Developed, trained, and fine-tuned AI models to predict customer demographics with 84% precision rate using XGBoost, Random Forests, and Bi-LSTM Neural Networks in Python (TensorFlow, scikit-learn) and R
- Deployed & implemented AI models into production environment using Docker, Airflow, and GitLab to automatically handle 600,000 predictions daily, increasing research and survey sampling efficiency by 30%
- Enhanced survey analysis efficiency by 50% through testing and validating data accuracy on 600,000 data points using Python and Excel to extract insight and identify accurate data source for survey analysis
- Built data pipelines to automate model validation, notifying stakeholders of model drifts signals

PROJECTS

Job Tracker AI Agent | *Python, Gen AI, Agentic AI, LLMs, RESTful APIs*

September 2025 – October 2025

- Built an autonomous AI Agent leveraging ChatGPT LLMs, Gmail API, and Google Sheets API to track, analyze, and update job application data from emails in real time, capable of handling 1,000+ emails daily
- Engineered a semantic extraction pipeline achieving 100% accuracy in identifying company, role, recruiter, and application status details across diverse email formats
- Reduced manual email tracking time by 95% through automating end-to-end CRM synchronization, enabling 24/7, zero-maintenance system performance

Customer Demographics Prediction - SMUD | *Python, R, Docker, Airflow*

December 2024 – August 2025

- Optimized model accuracy by 10% through hyperparameter tuning using Bayesian Optimization and AutoML
- Performed feature engineering by developing a K-means clustering pipeline to enhanced model's accuracy by 20%, processing over 7,000 geospatial records using ArcGIS, shapely, and Matplotlib
- Improved data quality by 100% through missing data imputation using MICE and k-NN, ensuring robust analysis

LEADERSHIP & ACTIVITIES

Head of Career Development & Alumni Relations

Davis Economics Collective (DEC)

September 2024 – June 2025

Davis, CA

- Led the Career Development & Alumni Relations team organizing 12 professional workshops with 100% retention rate, enhancing member engagement and professional growth
- Established and maintained relationships with over 60 economics alumni from Adobe, NVIDIA, etc. to support DEC initiatives and foster a robust professional network

PROFESSIONAL CERTIFICATES

Docker Foundations Professional Certificate

May 2025

IBM Python for Data Science, AI & Development

June 2024

IBM Data Visualization and Dashboards with Excel and Cognos

May 2024